

Reproduction of a Research Claim from Horowitz (2018), from American Sociological Review

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Last Updated 2020-10-29

Claim Summary.

The claim selected for reproduction from Horowitz (2018) is that while individuals with college experience use verbal skills more than those without a college background do, this advantage declines as the proportion of college-educated individuals increases. This reflects the following from the paper's abstract: "Higher-education expansion erodes the value of a college degree, and college-educated workers are at greater risk for underemployment in less cognitively demanding occupations." The claim is tested by regressing skill utilization (derived from O*NET occupational skill ratings) on individuals' education levels, relative education in their birth cohort and region net of age, period, cohort, and other demographic trends. The key independent variables are three dummy variables for educational attainment (completed some college, completed a four-year degree, or earned more education than a four-year degree, with individuals who have completed no college as the reference category), the proportion of individuals with a college degree, and the interaction of this variable with each of the dummy variables for education level (see Table 1 for full list of variables for the full model). For the purposes of the SCORE project, the effect of interest is the interaction between four-year college completion and the college proportion on use of verbal skills for men.

Replication Criteria.

A successful replication of the key SCORE test would show that, among men, the interaction between four-year college completion and the proportion of individuals with a college degree on use of verbal skills will be negative and significant.

Replication Result.

The (very long) regression table for the key test of interest is shown below. The key coefficient is for `degree_proportion x EDUC_RECDEBA Degree`. This table can also be found [here](#).

```
modelsummary(m,  
  statistic = 'std.error',  
  stars = TRUE)
```

Model 1

	Model 1
(Intercept)	2.722** (1.148)
degree_proportion	0.217*** (0.041)
EDUC_RECODESome College, No Degree	0.677*** (0.013)
EDUC_RECODEBA Degree	1.031*** (0.014)
EDUC_RECODEMore than 4-year Degree	1.244*** (0.017)
AGE	0.024*** (0.004)
I(AGE^2)	- 0.000*** (0.000)
I(AGE^3)	0.000 (0.000)
cohort51	0.077 (0.170)
cohort52	0.057 (0.167)
cohort53	0.122 (0.167)
cohort54	0.194 (0.167)
cohort55	0.143 (0.167)
cohort56	0.178 (0.167)
cohort57	0.167 (0.167)
cohort58	0.132 (0.167)
cohort59	0.178 (0.167)
cohort510	0.211

	Model 1
	(0.168)
cohort511	0.242
	(0.168)
cohort512	0.279*
	(0.169)
cohort513	0.293*
	(0.169)
cohort514	0.294*
	(0.170)
YEAR	-0.000
	(0.001)
MARST_RECODENever Married	-0.361
	(0.231)
MARST_RECDEWidowed / Divorced / Separated	-0.090
	(0.142)
RACE_RECDEHispanic	-
	0.136***
	(0.048)
RACE_RECDEOther	0.205
	(0.425)
RACE_RECDEWhite	0.539***
	(0.170)
METRONot in metro area	-
	0.082***
	(0.002)
METROCentral city	0.010***
	(0.002)
METROOutside central city	0.053***
	(0.002)
REGIONMiddle Atlantic Division	1.430**
	(0.645)
REGIONEast North Central Division	0.454
	(0.628)
REGIONWest North Central Division	2.135***
	(0.665)

	Model 1
REGIONSouth Atlantic Division	3.219*** (0.613)
REGIONEast South Central Division	2.392*** (0.820)
REGIONWest South Central Division	1.393** (0.707)
REGIONMountain Division	-0.006 (0.678)
REGIONPacific Division	0.369 (0.633)
degree_proportion × EDUC_RECODESome College, No Degree	- 0.495*** (0.036)
degree_proportion × EDUC_RECODEBA Degree	0.088** (0.038)
degree_proportion × EDUC_RECODEMore than 4-year Degree	0.449*** (0.046)
cohort51 × RACE_RECODEHispanic	-0.999* (0.558)
cohort52 × RACE_RECODEHispanic	0.303 (0.203)
cohort53 × RACE_RECODEHispanic	0.357** (0.178)
cohort54 × RACE_RECODEHispanic	0.060 (0.117)
cohort55 × RACE_RECODEHispanic	0.027 (0.090)
cohort56 × RACE_RECODEHispanic	-0.028 (0.074)
cohort57 × RACE_RECODEHispanic	0.033 (0.063)
cohort58 × RACE_RECODEHispanic	0.188*** (0.060)
cohort59 × RACE_RECODEHispanic	0.046

	Model 1
	(0.057)
cohort510 × RACE_RECDEHispanic	0.051
	(0.056)
cohort511 × RACE_RECDEHispanic	-0.011
	(0.056)
cohort512 × RACE_RECDEHispanic	0.054
	(0.059)
cohort513 × RACE_RECDEHispanic	-0.080
	(0.061)
cohort514 × RACE_RECDEHispanic	NA
cohort51 × RACE_RECDEOther	-0.091
	(0.434)
cohort52 × RACE_RECDEOther	0.052
	(0.427)
cohort53 × RACE_RECDEOther	0.002
	(0.426)
cohort54 × RACE_RECDEOther	-0.084
	(0.426)
cohort55 × RACE_RECDEOther	-0.056
	(0.425)
cohort56 × RACE_RECDEOther	-0.119
	(0.425)
cohort57 × RACE_RECDEOther	-0.125
	(0.425)
cohort58 × RACE_RECDEOther	-0.109
	(0.425)
cohort59 × RACE_RECDEOther	-0.099
	(0.425)
cohort510 × RACE_RECDEOther	-0.098
	(0.425)
cohort511 × RACE_RECDEOther	-0.114
	(0.425)
cohort512 × RACE_RECDEOther	-0.069
	(0.425)
cohort513 × RACE_RECDEOther	-0.049
	(0.425)

	Model 1
cohort514 × RACE_RECDEOther	-0.061 (0.426)
cohort51 × RACE_RECDEWhite	-0.119 (0.174)
cohort52 × RACE_RECDEWhite	-0.139 (0.171)
cohort53 × RACE_RECDEWhite	-0.195 (0.171)
cohort54 × RACE_RECDEWhite	-0.289* (0.170)
cohort55 × RACE_RECDEWhite	-0.253 (0.170)
cohort56 × RACE_RECDEWhite	-0.321* (0.170)
cohort57 × RACE_RECDEWhite	-0.363** (0.170)
cohort58 × RACE_RECDEWhite	-0.363** (0.170)
cohort59 × RACE_RECDEWhite	-0.384** (0.170)
cohort510 × RACE_RECDEWhite	-0.415** (0.170)
cohort511 × RACE_RECDEWhite	- 0.462*** (0.170)
cohort512 × RACE_RECDEWhite	- 0.501*** (0.170)
cohort513 × RACE_RECDEWhite	- 0.539*** (0.170)
cohort514 × RACE_RECDEWhite	- 0.561*** (0.171)
cohort51 × MARST_RECDENever Married	0.109 (0.236)
cohort52 × MARST_RECDENever Married	0.165

	Model 1
	(0.232)
cohort53 × MARST_RECODENever Married	0.144
	(0.232)
cohort54 × MARST_RECODENever Married	0.172
	(0.232)
cohort55 × MARST_RECODENever Married	0.182
	(0.231)
cohort56 × MARST_RECODENever Married	0.215
	(0.231)
cohort57 × MARST_RECODENever Married	0.236
	(0.231)
cohort58 × MARST_RECODENever Married	0.247
	(0.231)
cohort59 × MARST_RECODENever Married	0.247
	(0.231)
cohort510 × MARST_RECODENever Married	0.253
	(0.231)
cohort511 × MARST_RECODENever Married	0.263
	(0.231)
cohort512 × MARST_RECODENever Married	0.268
	(0.231)
cohort513 × MARST_RECODENever Married	0.293
	(0.231)
cohort514 × MARST_RECODENever Married	0.318
	(0.231)
cohort51 × MARST_RECDEWidowed / Divorced / Separated	-0.128
	(0.147)
cohort52 × MARST_RECDEWidowed / Divorced / Separated	-0.019
	(0.144)
cohort53 × MARST_RECDEWidowed / Divorced / Separated	-0.040
	(0.143)
cohort54 × MARST_RECDEWidowed / Divorced / Separated	-0.020
	(0.143)

	Model 1
cohort55 × MARST_RECODEWidowed / Divorced / Separated	-0.007 (0.143)
cohort56 × MARST_RECODEWidowed / Divorced / Separated	0.018 (0.143)
cohort57 × MARST_RECODEWidowed / Divorced / Separated	0.006 (0.143)
cohort58 × MARST_RECODEWidowed / Divorced / Separated	0.023 (0.143)
cohort59 × MARST_RECODEWidowed / Divorced / Separated	-0.008 (0.143)
cohort510 × MARST_RECODEWidowed / Divorced / Separated	-0.011 (0.143)
cohort511 × MARST_RECODEWidowed / Divorced / Separated	0.011 (0.143)
cohort512 × MARST_RECODEWidowed / Divorced / Separated	-0.006 (0.143)
cohort513 × MARST_RECODEWidowed / Divorced / Separated	0.040 (0.143)
cohort514 × MARST_RECODEWidowed / Divorced / Separated	0.015 (0.145)
EDUC_RECODESome College, No Degree × REGIONMiddle Atlantic Division	-0.008 (0.009)
EDUC_RECODEBA Degree × REGIONMiddle Atlantic Division	0.002 (0.009)
EDUC_RECODEMore than 4-year Degree × REGIONMiddle Atlantic Division	0.036*** (0.010)

	Model 1
EDUC_RECODESome College, No Degree × REGIONEast North Central Division	- 0.069*** (0.009)
EDUC_RECODEBA Degree × REGIONEast North Central Division	0.076*** (0.009)
EDUC_RECODEMore than 4-year Degree × REGIONEast North Central Division	0.104*** (0.011)
EDUC_RECODESome College, No Degree × REGIONWest North Central Division	- 0.139*** (0.009)
EDUC_RECODEBA Degree × REGIONWest North Central Division	-0.022** (0.009)
EDUC_RECODEMore than 4-year Degree × REGIONWest North Central Division	0.053*** (0.012)
EDUC_RECODESome College, No Degree × REGIONSouth Atlantic Division	0.024*** (0.008)
EDUC_RECODEBA Degree × REGIONSouth Atlantic Division	0.051*** (0.008)
EDUC_RECODEMore than 4-year Degree × REGIONSouth Atlantic Division	0.056*** (0.010)
EDUC_RECODESome College, No Degree × REGIONEast South Central Division	- 0.065*** (0.012)
EDUC_RECODEBA Degree × REGIONEast South Central Division	0.065*** (0.012)
EDUC_RECODEMore than 4-year Degree × REGIONEast South Central Division	0.159*** (0.016)
EDUC_RECODESome College, No Degree × REGIONWest South Central Division	0.000 (0.010)

	Model 1
EDUC_RECODEBA Degree × REGIONWest South Central Division	0.107*** (0.010)
EDUC_RECODEMore than 4-year Degree × REGIONWest South Central Division	0.128*** (0.013)
EDUC_RECODESome College, No Degree × REGIONMountain Division	- 0.083*** (0.009)
EDUC_RECODEBA Degree × REGIONMountain Division	0.002 (0.009)
EDUC_RECODEMore than 4-year Degree × REGIONMountain Division	0.036*** (0.012)
EDUC_RECODESome College, No Degree × REGIONPacific Division	-0.004 (0.008)
EDUC_RECODEBA Degree × REGIONPacific Division	0.055*** (0.008)
EDUC_RECODEMore than 4-year Degree × REGIONPacific Division	0.082*** (0.010)
YEAR × REGIONMiddle Atlantic Division	-0.001** (0.000)
YEAR × REGIONEast North Central Division	-0.000 (0.000)
YEAR × REGIONWest North Central Division	- 0.001*** (0.000)
YEAR × REGIONSouth Atlantic Division	- 0.002*** (0.000)
YEAR × REGIONEast South Central Division	- 0.001*** (0.000)
YEAR × REGIONWest South Central Division	-0.001** (0.000)

	Model 1
YEAR × REGIONMountain Division	-0.000 (0.000)
YEAR × REGIONPacific Division	-0.000 (0.000)
Num.Obs.	1189425
R2	0.326
R2 Adj.	0.326
AIC	2796653 .3
BIC	2798379 .7
Log.Lik.	- 1398182 .650
F	4044.11 4

* p < 0.1, ** p < 0.05, *** p < 0.01

The coefficient on the key interaction of interest, Degree Proportion x BA Degree, is positive: $\hat{\beta} = .088$, $t = .088/.038 = 2.316$, $p = .019$, based on a sample of 1,189,425 observations (which matches the preregistered sample size).

This fails to replicate the original result. Some features of the regression are similar to the original: the sign and significance of all of the non-interacted degree proportion and education terms shown in the above table, as well as the interaction between degree proportion and Some College, No Degree match, and the R^2 value of .326 is similar to the .339 in the original paper. However, the coefficient on the interaction term between degree proportion and BA degree is of an opposite sign to what was expected by the replication criteria.

The claim selected for the original study yielded a negative regression interaction term in the focal test, corresponding to an increased local degree proportion reducing the effect of holding a four-year degree on the use of verbal skills. The replication yielded a positive regression interaction term in the focal test, corresponding to an effect in a different pattern compared to the original study.

Deviations from preregistration.

There were no deviations from the preregistration plan, or changes to the preregistered script (except to remove the line limiting analysis to a random 5% subset of the data).

Deviations from the original study.

There are two major differences from the original study, and one minor difference, as described in the preregistration document.

The minor difference is that the original paper did not describe dropping people on the basis of retirement. This dropping of observations was in the replication code, but because it was not in the paper, I did not do it in this reproduction. This resulted in a difference of a few thousand observations.

The first major difference is that the original study did not describe how it matched Census occupation codes in Flood et al. (n.d.) with SOC occupation codes. It described using Autor and Dorn (2013) to harmonize Census occupation codes (although some details were omitted), but did not describe how the match between the Autor and Dorn (2013) codes and the SOC codes used in National Center for O*NET Development (n.d.) was performed, and information on this was not available in the original study's replication files. Differences in this procedure are a main reason why the observation numbers differ so sharply from the original study (by about 300,000).

In a [correction](#) to the original study, the author describes a change in the data processing code that increased the number of observations by a large amount, narrowing the gap in observation numbers to about 100,000. However, the author's correction did not find the difference in results found in this reproduction.

The second major difference is in the construction of the Verbal dependent variable, which is the result of a confirmatory factor analysis. The original study (and original study's replication code) did not describe how this variable was created beyond mentioning that it used confirmatory factor analysis and showing the path diagram used in that CFA. So it is highly unlikely that the CFA performed in this reproduction uses the exact same methods as the original study.

Description of materials provided.

The following materials are publicly available on the OSF project site:

<https://osf.io/6ye5m/files/>

- R script for compiling data together: **compile_data.R**
- Data files used to link Census occupation codes to occ1990dd codes to SOC occupation codes, from Autor and Dorn (2013) and Acemoglu and Autor (2011): **census-2000-1990dd-weights.dta**, **soc-census-weights.dta**, and the **occXXXX_occ1990dd.dta** files, where XXXX can be 1950, 1960, 1970, 1980, 1990, or 2000.
- R script for performing analysis: **analysis_code_horowitz.R**
- Regression table output: **horowitz_regression.html**
- Variable documentation files: **variable_documentation.html**, containing descriptions of all variables, **summary_table.html**, containing basic summary statistics, and **summary_table_no_METAREA.html**, which contains basic summary statistics of everything but the METAREA variable, allowing it to be considerably shorter

- **README.txt** describing how to use the above files other than **analysis_code_horowitz.R**

Not included in the OSF project site:

- Current Population Survey downloads from IPUMS (notes on creating an appropriate data file from IPUMS are in **compile_data.r** and **README.txt**). This requires an IPUMS account but is otherwise free.
- ONET Occupational task data (available at [ONET Database Releases Archive](#))

Citations.

Acemoglu, Daron, and David H. Autor. 2011. "Skills, Tasks and Technologies: Implications for Employment and Earnings." In *Handbook of Labor Economics*, 4:1043–1171. Elsevier.

Autor, David H., and David Dorn. 2013. "The Growth of Low-Skill Service Jobs and the Polarization of the Us Labor Market." *American Economic Review* 103 (5): 1553–97.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, and J. Robert Warren. n.d. "Integrated Public Use Microdata Series, Current Population Survey: Version 7.0 [Dataset]." Minneapolis, MN: IPUMS, 2020, <https://doi.org/10.18128/D030.V7.0>.

Horowitz, Jonathan. 2018. "Relative Education and the Advantage of a College Degree." *American Sociological Review* 83 (4): 771–801.

National Center for O*NET Development. n.d. "O*NET Database Releases Archive." O*NET Resource Center.